

LAVANYA SHANMUGARAJ

DATA ANALYST

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PROFESSIONAL SUMMARY

B.Sc. Information Technology graduate with hands-on experience in data analysis, data cleaning, validation, and reporting through internships and practical projects. Proficient in MS Excel, including Pivot Tables, VLOOKUP, SUM, SUMIFS, and IF functions, with working knowledge of SQL, Power BI, and Python. Experienced in analyzing retail transaction data, preparing structured reports, identifying data discrepancies, and presenting insights clearly. Detail-oriented and analytical, with strong problem-solving skills and a focus on data accuracy and timely reporting.

CORE SKILLS

- Data Analysis & Statistics:** EDA, Hypothesis Testing, Statistical Analysis, Data Cleaning & Validation, Data Accuracy & Validation, Sales Data Analysis, A/B-Style Comparison, Report Preparation & Documentation, Data Confidentiality & Integrity.
- Programming & Query:** SQL (MySQL, MS SQL Server), Python (Pandas, NumPy).
- BI & Data Visualization:** Power BI, Tableau, Excel (Pivot Tables, Formulas, VLOOKUP), Matplotlib, Dashboard Development, KPI Reporting.
- Professional Skills:** Problem Solving, Analytical Thinking, Time Management, Stakeholder Communication.

INTERNSHIP EXPERIENCE

Sysstan IT Solutions | Aug 2026 – Sept 2026

- Cleaned and validated railway schedule data, confirming zero nulls and duplicate records and verifying station sequencing for data accuracy.
- Analyzed journey durations and categorized 100% of routes into Short (<3 hours), Medium (3–6 hours), and Long (>6 hours).
- Identified high-traffic stations through frequency analysis (CST-Mumbai: 1,027 unique trains) and built pivot tables and cross-tabulations to analyze station-route distribution.
- Developed an interactive train enquiry tool to retrieve direct trains with departure/arrival times and estimated journey duration.

Data Analytics Virtual Internship — Quantum (Forage)

- Cleaned and engineered 260K+ retail transaction records to identify customer purchasing behaviour and segment-level spending patterns.
- Evaluated a retail trial across stores 77, 86, and 88 using matched control stores and statistical testing; identified significant sales increases of up to 62%.
- Identified customer acquisition as the primary driver of trial-period sales growth and developed customer-segment insights and recommendations for retention and targeted offers.

PROJECTS

Fake Job Posting Analysis — EDA & Fraud Pattern Dashboard

Tools: MySQL Workbench, Excel, Power BI

- Conducted EDA on 17,079 job postings, resolving nulls, duplicates, and formatting inconsistencies using Excel and MySQL Workbench.
- Identified 3 statistically verified fraud indicators: no company logo (16.21% fraud rate), part-time roles (9.83%), remote roles (8.66%).
- Uncovered a counter-intuitive finding — fraudulent postings listed a salary 25.12% of the time vs. 14.87% for genuine ones, indicating salary is used as bait.
- Built an interactive Power BI dashboard tracking fraud trends by fraud rate, location, industry, and employment type.

Instacart Retail Sales & Customer Insights

Tools: Excel, Python (Pandas), Power BI

- Analyzed 3.42M+ orders across 206K+ customers and 49K+ products using Python/Pandas to identify purchasing patterns, customer frequency, and reorder behavior.
- Segmented customers by purchase frequency, identifying High-Frequency customers with a 69.57% reorder rate and Medium-Frequency customers as the largest segment (46.43%).
- Developed Excel MIS reports and an interactive Power BI dashboard to track KPIs, department/product performance, reorder behavior, customer segments, and order timing.

EDUCATION | CERTIFICATION

- Bachelor of Science in Information Technology** — Fatima College, Madurai, Tamil Nadu
| June 2023 – April 2026 | CGPA: 8.1 |
- HSC (12th Grade)** — St. Joseph's Girls Higher Secondary School, Dindigul, Tamil Nadu
| June 2022 – April 2023 | PERCENTAGE: 88% |
- Data Visualization: Empowering Business with Effective Insights — Tata Group Job Simulation
- Data Science & Analytics — HP LIFE
- Data Analytics — IBM